

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 9: Superprompt-Critical Excursion Following Step Reactivity Insertion

STEP REACTIVITY INSERTION

Assumptions:

1. Prompt Kinetics Approximation (PKA) $\rightarrow$

$$\dot{p} = \frac{\rho - \beta}{\Lambda} p \quad p(0) = p^0 = \frac{\rho_1}{\rho_1 - \beta} p_0$$

2. Doppler feedback is described by an energy coefficient γ (temperature independent) $\rightarrow$

$$C_{QT} \left(\frac{\partial \rho}{\partial T} \right)_{Dop} = \gamma_e^D \left[\frac{\delta \rho}{MW \cdot s} \right] = \text{energy coefficient}$$

3. Temperature rise by the linear energy model (adiabatic BCs) $\rightarrow \quad \rho(t) = \rho_0 + \gamma \int_0^t p(t') dt'.$

4. Neglected stationary cooling $\rightarrow \quad \gamma \int_0^t (-p_0) dt' = \gamma p_0 t = 0$

5. Prompt reactivity $\rho_p = (\rho - \beta)$ is defined as “reactivity above β ” $\rightarrow \quad \rho - \beta = \rho_p$

STEP REACTIVITY INSERTION

The prompt kinetics equation then assumes the form:

$$\dot{p} = \frac{\rho_p}{\Lambda} p \quad \text{and} \quad p(0) = p^0 = \frac{\rho_1}{\rho_{p1}} p_0$$

After the onset of the transient:

- Flux rises rapidly
- Negative feedback → reactivity decreases: at $t = t_m$, the decreasing reactivity becomes equal to β , i.e., $\rho_p = 0$
- prompt reactivity vanishes at the same time when the flux has a maximum:

$$\rho_p(t_m) = 0 \quad \rightarrow \quad \rho(t_m) = \beta \quad \text{or} \quad -\gamma \int_0^{t_m} p(t') dt' = \rho_1 - \beta$$

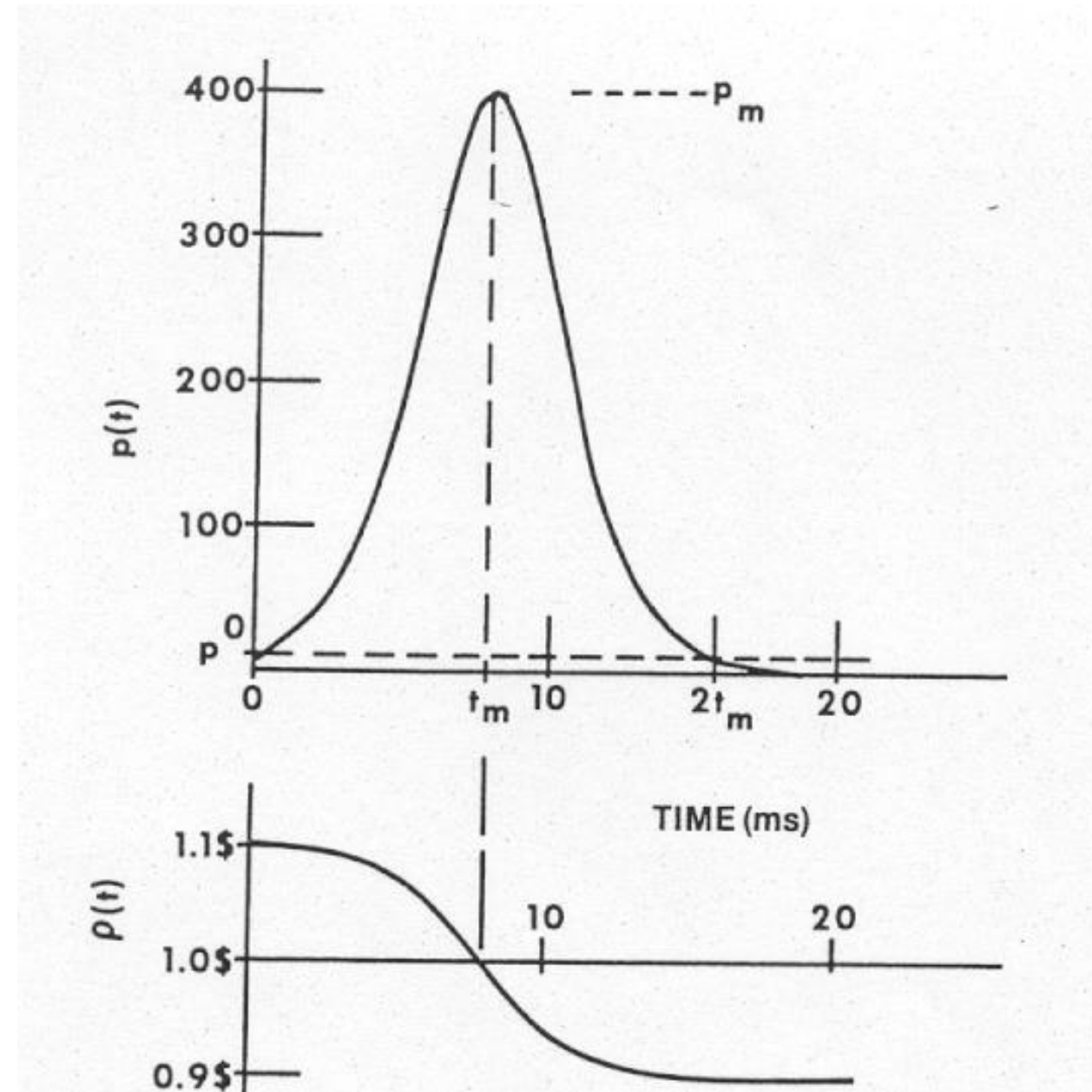
Consequently, the prompt part of the inserted reactivity is just compensated when the flux has reached its maximum.

STEP REACTIVITY INSERTION

Superprompt-critical transient

$$\rho_1 = 1.1\%$$

$$p_0 = 0.1p_n$$



STEP REACTIVITY INSERTION

The energy release during the first part of the transient, i.e., from the beginning to the maximum of the excursion is then:

$$\Delta Q_m = \int_0^{t_m} P(t') dt' = -\frac{\rho_{p1}}{\gamma_e}$$

The energy release is independent of the neutron generation time

We obtain the corresponding feedback coefficients as:

$$\frac{\Delta \rho}{\Delta Q} = \frac{\partial \rho}{\partial Q} = \gamma_e = \text{energy coefficient} \quad \text{and} \quad \frac{\Delta \rho}{\Delta T} = \frac{\partial \rho}{\partial T} = \gamma_T = \text{temperature coefficient}$$

These equations hold at all times - at the maximum of the transient $\Delta \rho = -\rho_{p1}$.

$$\text{Then, } \Delta Q_m = -\frac{\rho_{p1}}{\gamma_e} \quad \text{and} \quad \Delta T_m = -\rho_{p1} \left(\frac{\partial \rho}{\partial T} \right)^{-1}.$$

The rate of the reactivity reduction by feedback is largest at the power maximum: $\dot{\rho}_m = \gamma_e P_m$.

Thus, the reactivity quickly passes through the value of β (and higher) into the range below prompt critical.

STEP REACTIVITY INSERTION

The applicability of the prompt kinetics approximation (PKA) below $\rho = \beta$ is questionable.

When the delayed neutron source determines the flux level (through source multiplication):

→ PKA gives unrealistic results (neglects delayed neutrons).

→ If $\rho < \beta$, PKA only describes the prompt die-away of an existing neutron population.

Before the maximum of such transients, it is the die-away of the large neutron population built up → PKA yields quite accurate results during this phase of the transient.

During this phase, the flux level is reduced from its maximum value to a level determined by source multiplication of the delayed neutrons.

After this phase PKA is not applicable.

STEP REACTIVITY INSERTION

In conclusion, the applicability of the PKA model is normally **restricted** to a reactivity range above β .

Exception: PKA is applicable to the range $\rho < \beta$ if $\rho(t)$ enters the range quickly from the superprompt domain, because then the transient consists merely of the die-away describable by the PKA until finally the delayed neutrons take over.

STEP REACTIVITY INSERTION

Investigation of the First Integral

The consideration of the differential equation for superprompt-critical transients with linear energy feedback has already provided important information about:

- *the reactivity,*
- *energy release,*
- *and temperature rise between the onset and the maximum of the excursion*

Further information is obtained by considering the "**first integral**" of the original equation, which can be brought in the form of a differential equation of second order

The first integral also provides information on the total reactivity feedback during the transient

STEP REACTIVITY INSERTION

$$\dot{p} = \frac{\rho - \beta}{\Lambda} p$$

$$\rho(t) = \rho_1 + \gamma \underbrace{\int_0^t p(t') dt'}_{=q}$$

$$\dot{q} = p$$

$$\ddot{q} = \dot{p}$$

rewritten into a second-order differential equation $\rightarrow$

$$\Lambda \ddot{q} = (\rho_{p1} + \gamma q) \dot{q}$$

$$q(t) = \int_0^t p(t') dt'$$

Introducing the prompt reactivity as the unknown function

$$\dot{\rho}_p = \gamma p = \gamma \dot{q} \quad \ddot{\rho}_p = \gamma \dot{p} = \gamma \ddot{q}$$

$$\longrightarrow \quad \Lambda \ddot{\rho}_p = \rho_p \dot{\rho}_p \quad \rightarrow \text{a nonlinear second-order differential equation}$$

STEP REACTIVITY INSERTION

The first integral is readily found to be $\Lambda \int_0^t \frac{d\dot{\rho}_p}{dt'} dt' = \int_0^t \rho_p \frac{d\rho_p}{dt'} dt'$

This gives a first integral in the form of $\Lambda[\dot{\rho}_p(t) - \dot{\rho}_p(0)] = \frac{1}{2}[\rho_p^2(t) - \rho_p^2(0)]$

The initial values are given by $\rho_p(0) = \rho_{p1} = \rho_p - \beta$ and $\dot{\rho}_p(0) = \gamma p^0$

Thus, $\Lambda[\dot{\rho}_p(t) - \gamma p^0] = \frac{1}{2}[\rho_p^2(t) - \rho_{p1}^2]$

$$\Lambda\gamma[p(t) - p^0] = \frac{1}{2}[\rho_p^2(t) - \rho_{p1}^2]$$

STEP REACTIVITY INSERTION

At the time of the maximum of the transient:

$$\left. \begin{aligned} \Lambda\gamma[p(t) - p^0] &= \frac{1}{2}[\rho_p^2(t) - \rho_{p1}^2] \\ \rho_p &= 0 \end{aligned} \right\} \Lambda\gamma(p_m - p^0) = -\frac{\rho_{p1}^2}{2}$$

This gives a formula for the maximum flux amplitude or power of the excursion: $p_m = p^0 - \frac{\rho_{p1}^2}{2\Lambda\gamma}$

The maximum power rise in megawatts with an energy coefficient γ_e [\$/MW·s] is:

$$P_m - P^0 = -\frac{\rho_{p1}^2}{2\Lambda\gamma_e}$$

Note: The power rise $P_m - P^0$ is inversely proportional to the generation time and the energy coefficient, and proportional to ρ_{p1}^2

If the temperature dependence of the energy coefficient is neglected, the power rise is independent of the initial power

STEP REACTIVITY INSERTION

Consider, for example, the transient terminated when $p(t)$ has returned to the pseudo-initial flux, p^0 , say at $t = t_2$

Then, the left side of $\Lambda\gamma[p(t) - p^0] = \frac{1}{2}[\rho_p^2(t) - \rho_{p1}^2]$

vanishes initially as well as when $p(t)$ is passing through p^0 at t_2 , i.e., at the "end" of the transient

Thus,

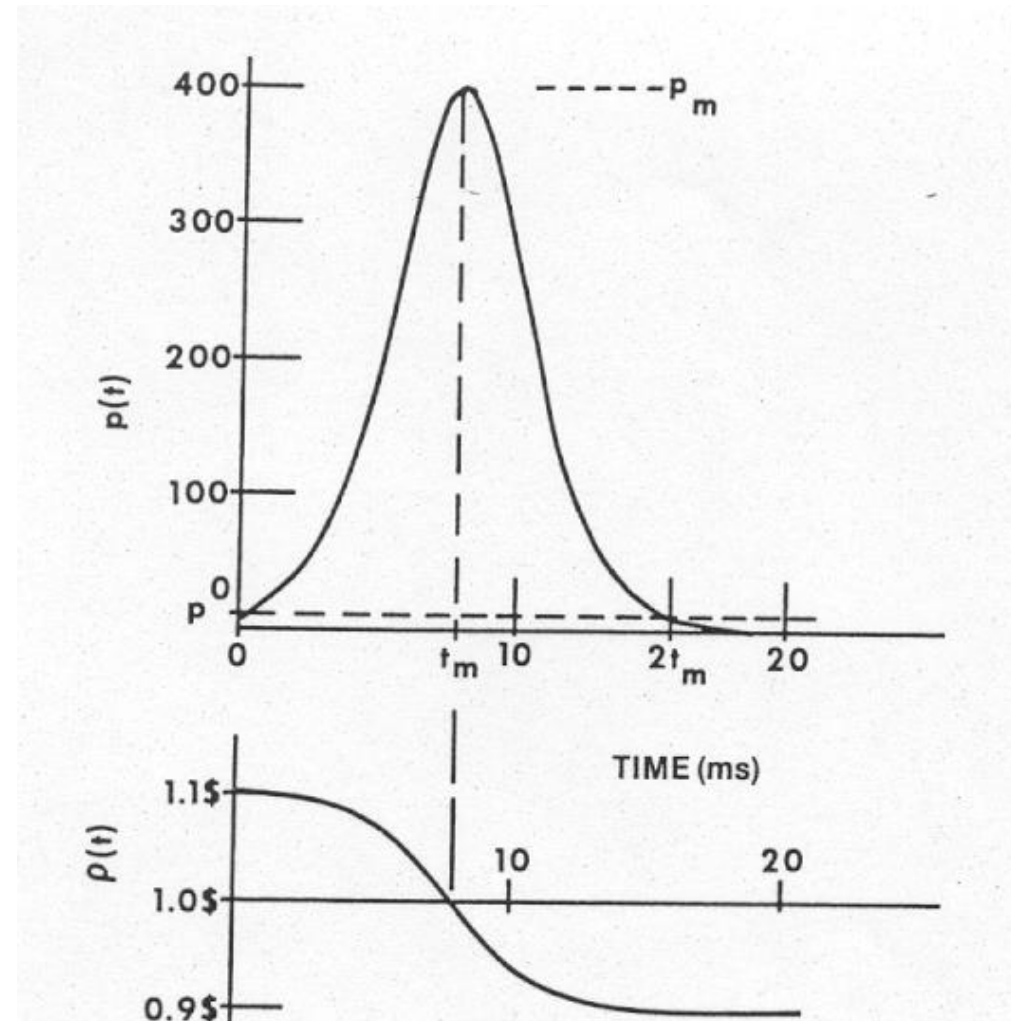
$$\rho_p^2(t) = \rho_{p1}^2 \text{ for } t = 0 \text{ and } t_2$$

with the two solutions: $\rho_p(0) = +\rho_{p1}$ at $t = 0$

$$\rho_p(t_2) = -\rho_{p1} \text{ at } t = t_2$$

STEP REACTIVITY INSERTION

The prompt feedback compensates the prompt reactivity twice during the transient - in other words, the reactivity swing is just two times the initial prompt reactivity



Superprompt-critical transient ($\rho_1 = 1.1\$, p_0 = 0.1p_n$) from the PKA with linear energy feedback model

STEP REACTIVITY INSERTION

In conclusion, the total energy release during a superprompt-critical transient is twice the energy release up to the maximum of the transient:

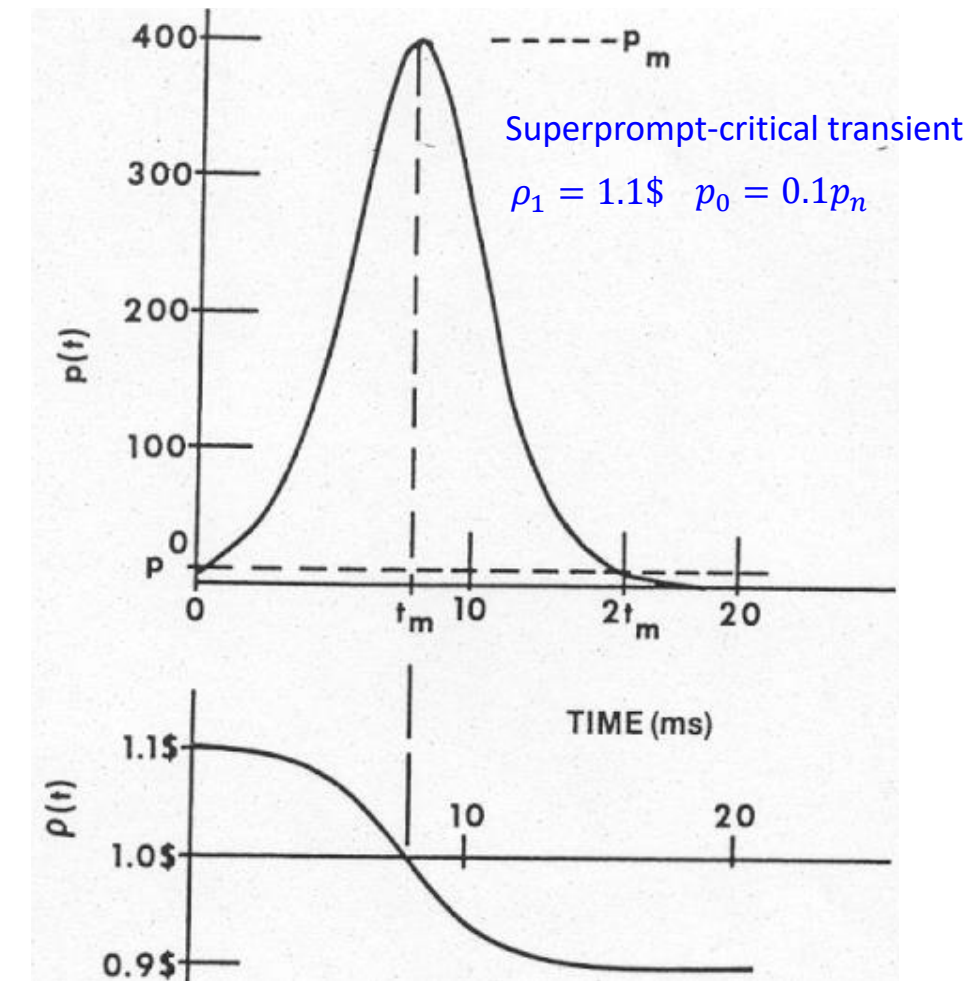
$$\Delta Q(t_2) = 2\Delta Q_m = -2 \frac{\rho_{p1}}{\gamma_e}$$

Thus, the total energy release during a superprompt-critical transient is independent of the generation time Λ , as it is for delayed supercritical transients

STEP REACTIVITY INSERTION

In summary:

1. The prompt reactivity vanishes at the same time when the flux has a maximum.
2. Consequently, the prompt part of the inserted reactivity is just compensated when the flux has reached its maximum.
3. The energy release is independent of the neutron generation time.
4. Below $\rho = \beta$, the PKA only describes the prompt die-away of an existing neutron population - it cannot account for the behavior of the delayed neutron source, except in the initial condition.
5. The prompt feedback compensates the prompt reactivity twice during the transient - in other words, the reactivity swing is just two times the initial prompt reactivity.

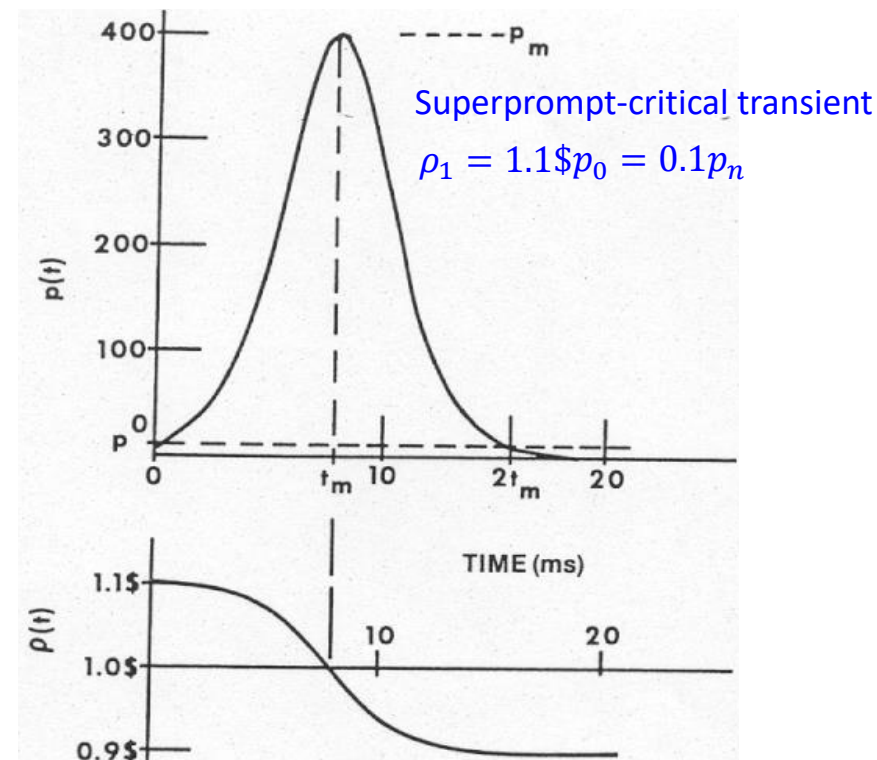


FLUX TRANSIENT DURING A SUPERPROMPT-CRITICAL EXCURSION

It can be shown that the PKA together with the linear energy feedback model describes the flux transient during a superprompt-critical excursion by:

$$p(t) = \frac{p_m}{\left\{ \cosh \left[\frac{\rho_b}{2\Lambda} (t - t_m) \right] \right\}^2} \quad \text{with maximum flux } p_m = -\frac{\rho_b^2}{2\Lambda\gamma}$$

Thus, the PKA together with the linear energy feedback model gives a transient that is symmetrical around $(t - t_m)$.



FLUX TRANSIENT DURING A SUPERPROMPT-CRITICAL EXCURSION

Burst width:

The "width" of the flux pulse (or burst), $\bar{\Delta}t$, is given as:

$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \left(1 - \frac{p^0}{p_m} \right)$$

Case 1:

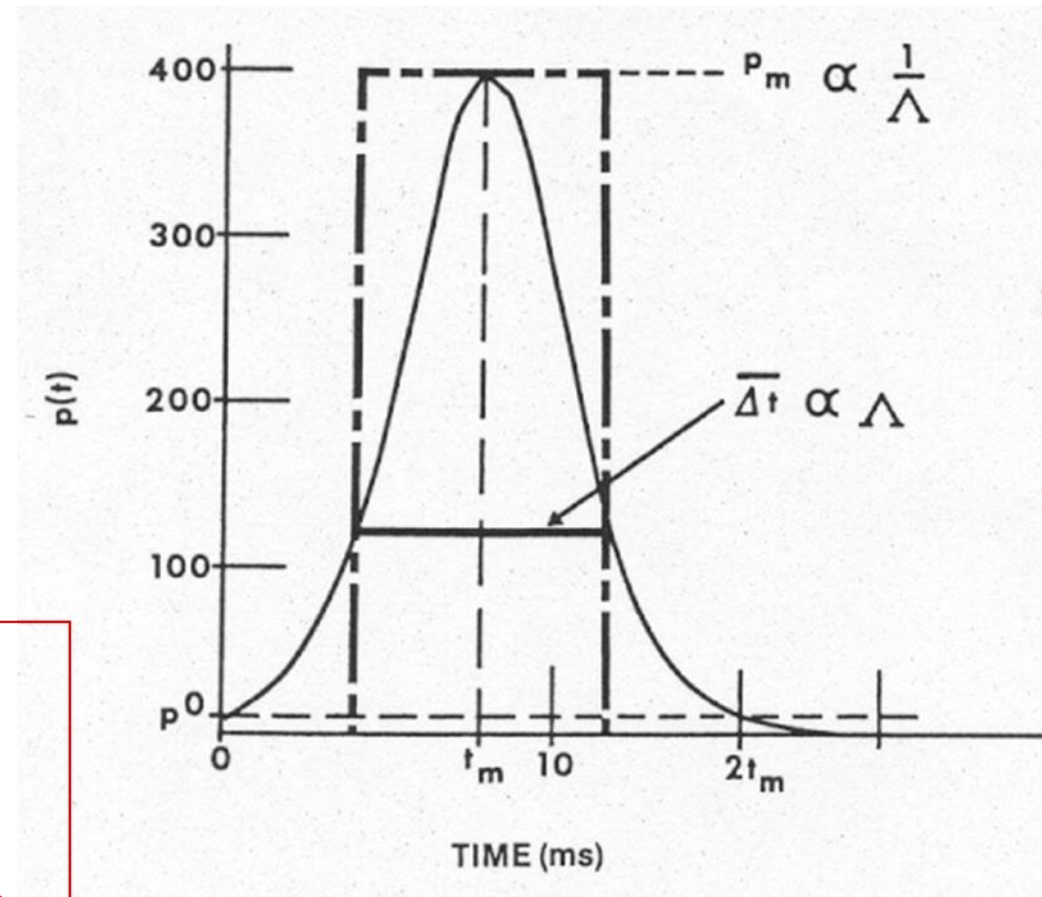
$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \quad \text{for } p_m \gg p^0$$

Λ -independent energy production

Case 2:

$$t_m \approx \frac{\Lambda}{\rho_{p1}} \left(\ln 4 \frac{p_m}{p^0} \right) \quad \text{for } p^0 < p_m$$

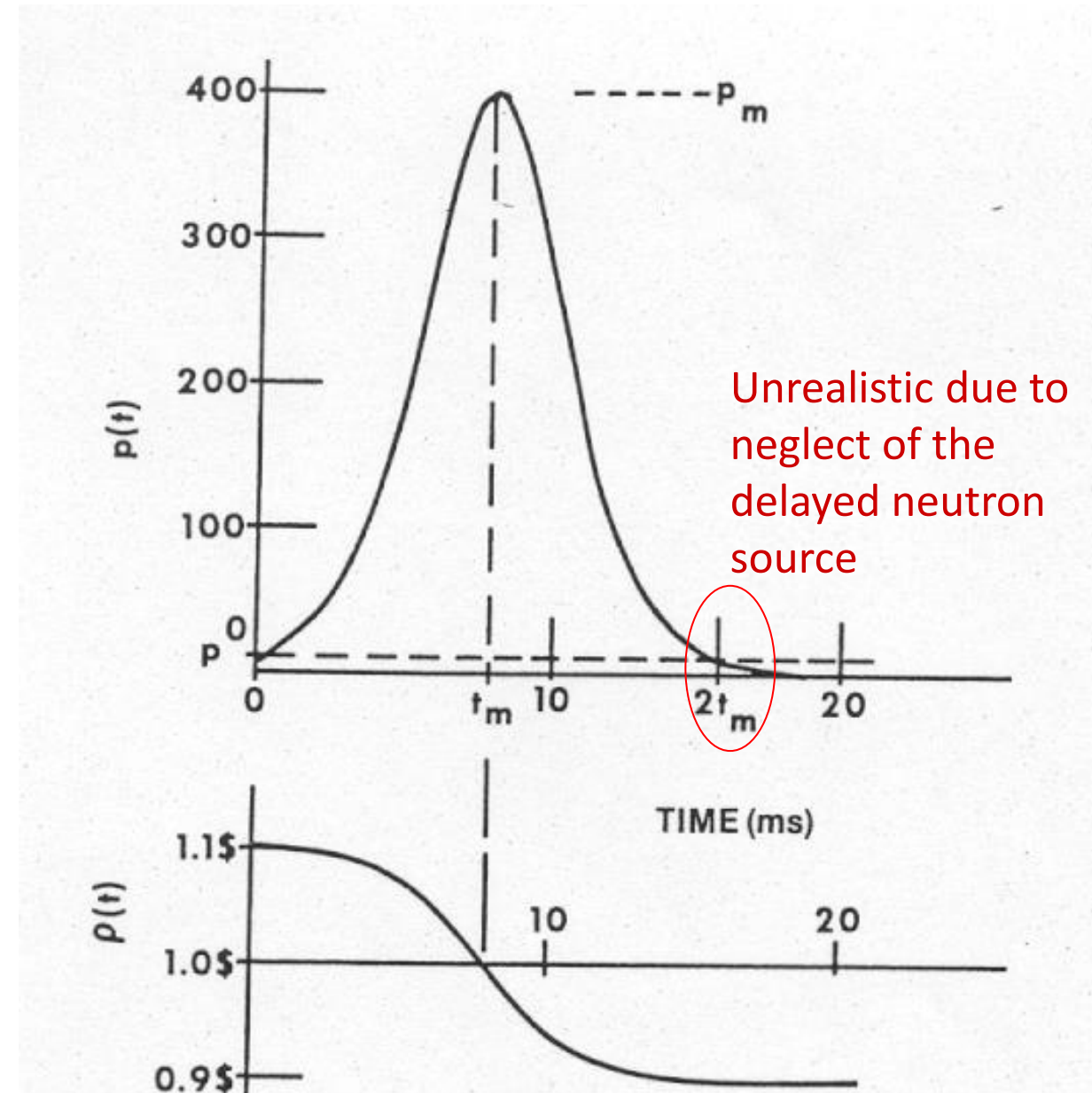
The maximum is quickly reached after the onset of the burst



POST-BURST FLUX TRANSIENT

The neutron flux decreases rapidly after it passes through its maximum.

$$p(t) = \frac{p_m}{\left\{ \cosh \left[\frac{\rho_b}{2\Lambda} (t - t_m) \right] \right\}^2}$$



POST-BURST FLUX TRANSIENT

- After the rapid flux changes during the burst are terminated, the further changes of the flux are slow so that $\Lambda \dot{p}$ can be neglected (PJA).
- An estimate of the flux level can then be obtained by estimating the delayed neutron source and applying the time-dependent source multiplication formula.
- A first approximation is obtained by using the initial value of the delayed neutron source and the reactivity after the burst:

$$\rho(t_2) = \beta - \rho_{p1}$$

- From the source multiplication consideration of the delayed neutrons, it follows that the flux is “held up” at least at a level (after the burst) given by:

$$p_{ab} \approx \frac{S_{d0}}{\beta - \rho} = \frac{S_{d0}}{\rho_{p1}} = \frac{\beta}{\rho_{p1}} p_0$$

POST-BURST FLUX TRANSIENT

- An improved estimate of the delayed neutron source multiplication requires accounting for the increase of the precursor population during the flux burst.
- During the short time of the burst, the $\bar{\lambda}$ kinetics approximation is applicable.
- Since the time interval considered is very small, the precursor decay can be neglected in the treatment of the precursor balance ($\bar{\lambda} \ll 1$).

$$\zeta(t_2) \approx \zeta_0 + \beta \int_0^{t_2} p(t') dt'$$

$$\bar{\lambda} \zeta_0 = \beta p_0$$

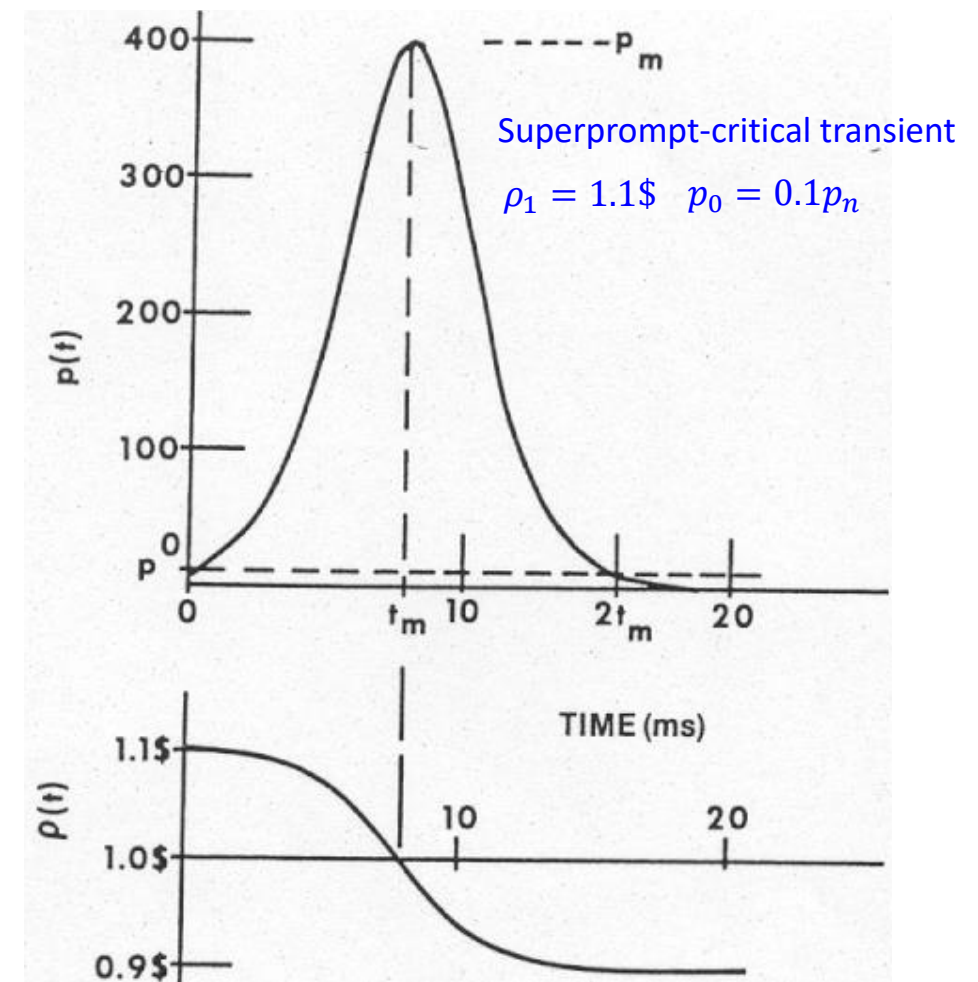
$$\bar{\lambda} \zeta(t_2) = \beta p_0 \left(1 - 2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} \right)$$

POST-BURST FLUX TRANSIENT

How much is the delayed neutron source increased?

$$\bar{\lambda}\zeta(t_2) = \beta p_0 \left(1 - 2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} \right)$$

$$2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} = \frac{-2(0.1\%) \left(0.565 \frac{1}{s} \right)}{(0.1) \left(-0.8 \frac{\%}{s} \right)} \approx 1.41$$



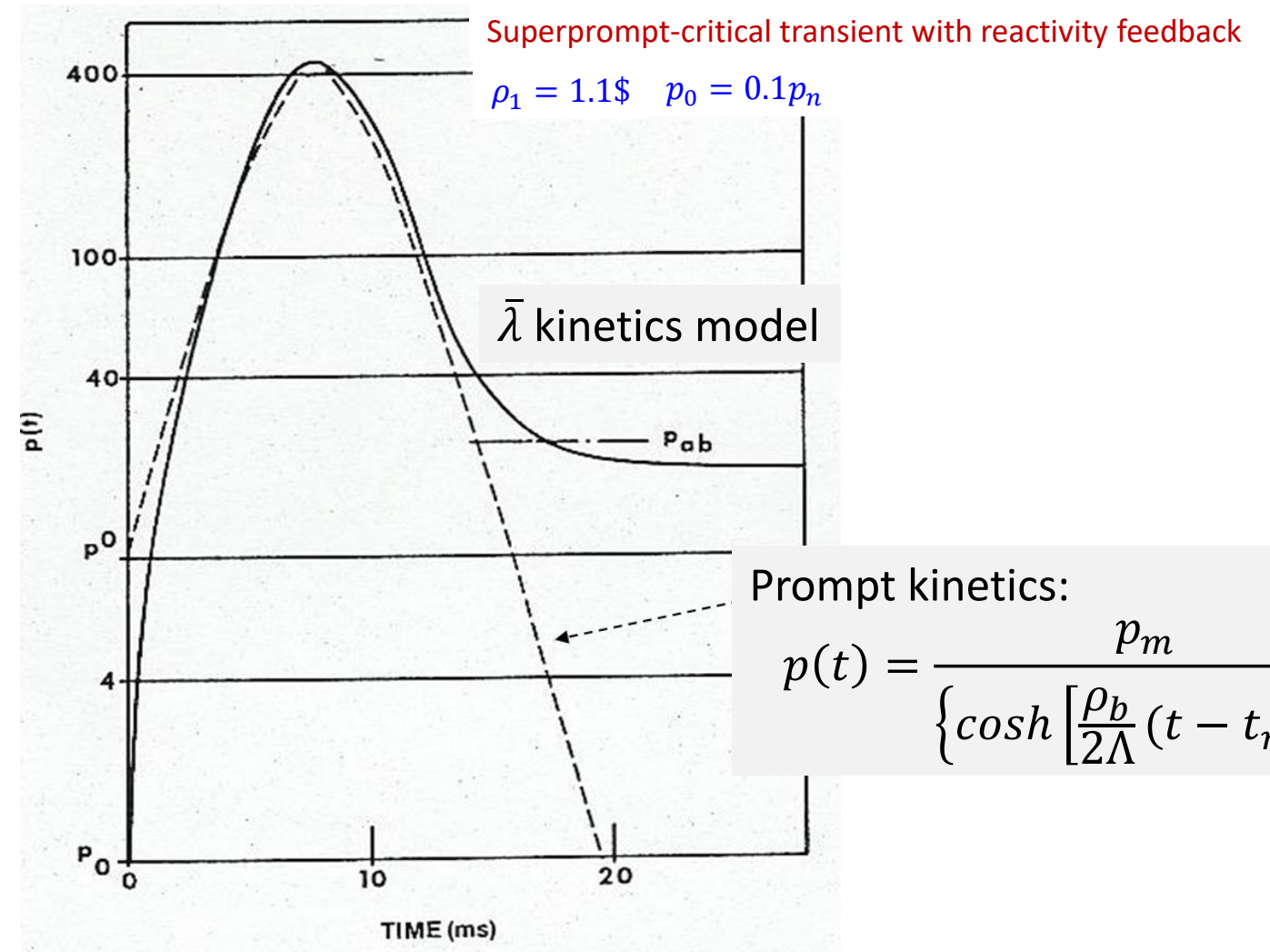
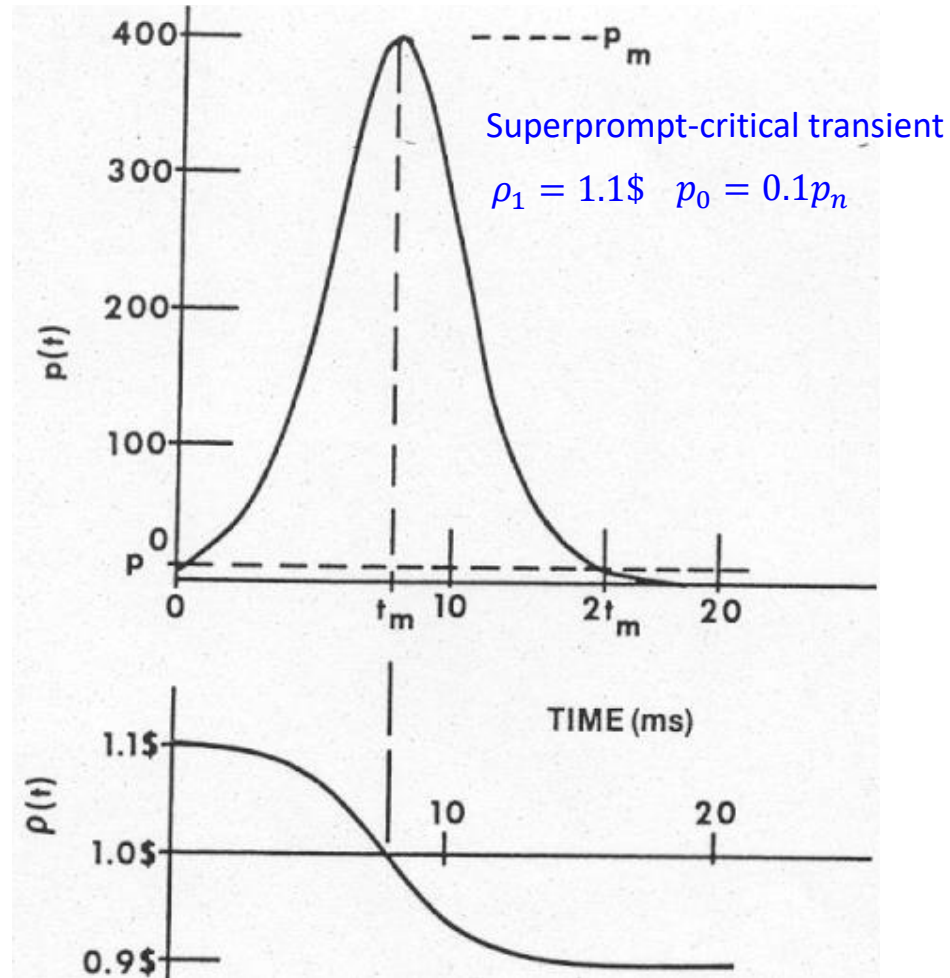
If the initial power was very low, the fraction of the precursors produced during the flux burst would be even larger.

The flux after the burst with the precursor increase included is given by:

$$p_{ab} = \frac{\beta p_0}{\rho_{p1}} \left(1 - 2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} \right)$$

POST-BURST FLUX TRANSIENT

Superprompt-critical transient with reactivity feedback



Prompt kinetics:

$$p(t) = \frac{p_m}{\left\{ \cosh \left[\frac{\rho_b}{2\Lambda} (t - t_m) \right] \right\}^2}$$

POST-BURST FLUX TRANSIENT

The further progression of the excursion is essentially the same as in a transient following a subprompt-critical step reactivity insertion.



The reactor power after the prompt burst is much larger than the nominal power.



This causes a further increase in the fuel temperature and a corresponding decrease in the reactivity due to the negative energy coefficient.



Eventually, the reactivity is reduced to zero and the flux approaches an asymptotic level above the initial flux.

POST-BURST FLUX TRANSIENT

The temperature increase during the entire transient is composed of:

1. The part produced during the prompt burst
- &
2. The part produced during the subsequent relatively gradual power decrease:

$$\Delta T_{total} = \Delta T_{burst} + \Delta T_{post-burst}$$
$$\Delta T_{total} = -\frac{1}{\gamma_T} [2(\rho_1 - \beta)] - \frac{1}{\gamma_T} [\rho_1 - 2(\rho_1 - \beta)] = -\frac{\rho_1}{\gamma_T}$$

In reactor accident analysis, ρ_1 is almost never much larger than β .

Then, only the small part of the temperature rise occurs during the prompt burst.

The larger part of the energy is released during the subprompt-critical phase of the excursion, before or after a prompt burst.

For example, for $\rho_1 = 1.1\%$, ~ 18% of the energy is released during the prompt burst and ~72% during the slower phase of the transient.

Next Class

*Superprompt-Critical Excursion Following Ramp Reactivity
Insertion*